实 验 报 告

课程名称：OpenGauss DBMS **实验编号：三**

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指导老师：火克华 **实验日期：**2024/4/5 **学生电话：18717024225**

一. 实验名称：

Lab3:OpenGauss SQL 语言基础查询(1)

二. 实验内容

1. OpenGauss SQL 语言基础

1.1. DQL:数据查询语言

A. select:用于查询数据库对象中的数据；
B. 选择：筛选符合条件的行，过滤掉不符合条件的行使其不显示或不输出；
C. 投影：筛选符合条件的列，过滤掉不符合条件的行使其不显示或不输出；
D. 连接：把数据存储在多个表里面，通过连接查询获得完整的数据信息。
E. 条件查询：
   1) 比较运算符（= != > < >= <= (IS NULL) in (BETWEEN...AND...)）
   2) 逻辑运算符（and or not ）
   3) 用 in 比较限定运算符
   4) 条件查询的结果用 order by 排序
F. 模糊查询：模糊查询使用 like 关键字和通配符结合来实现，通配符具体含义如下：
   1) %：代表匹配 0 个字符、1 个字符或多个字符。
   2) _：代表匹配有且只有 1 个字符。
   3) []：代表匹配范围内
   4) [^]：代表匹配不在范围内
   1) 聚合函数
      a) 数量 count()
      b) 最大值 max()
      c) 最小值 min()
      d) 求和 sum()
      e) 平均值 avg()
   2) 其它函数
      a) length(PeopleName)
      b) year(PeopleBirth),month(PeopleBirth),day(PeopleBirth)
      c) extract(year from PeopleBirth),extract(year from now()
      d) now()
      e) getdate()
      f) substr()

1.2. DCL 数据控制语言（体会）

A. grant:分配权限给数据库用户或角色;
B. revoke:回收数据库中用户或角色的权限;

三. 实验过程

1. 打开数据库 DB_em，进入如下操作:

   在 people 表中，根据指定列（姓名，性别，月薪，电话）查询性别为女的员工信息,并自定义中文列名:

```
ben_ben=# select peoplename 姓名,peoplesex 性别,peoplesalary 月薪,peoplephone 电话 from people where peoplesex = '女';
```

```
ben_ben=# select peoplename 姓名,peoplesex 性别,peoplesalary 月薪,peoplephone 电话 from people where peoplesex = '女';
   姓名    | 性别 |   月薪    |     电话
----------+------+----------+-------------
 周小顺    | 女   | 26000.00 | 13256854578
 钱小旦    | 女   | 13000.00 | 13985745871
 胡瑗瑗    | 女   | 17500.00 | 13437100050
 宋可晴    | 女   | 16000.00 | 13889562354
 李萌      | 女   | 10000.00 | 13698545841
 叶子飞云  | 女   | 12000.00 | 13558745874
 吴若欣    | 女   | 14000.00 | 13985478512
 刘婷      | 女   | 14500.00 | 13778787874
 王萌      | 女   |  9500.00 | 13352197364
 陆子虹    | 女   |  8500.00 | 13025457392
 刘星宇    | 女   | 18000.00 | 18717024225
(11 rows)
```

2. 查询月薪大于等于 10000 的员工信息(单条件)

```
ben_ben=# select * from people where peoplesalary >= 10000;
```

```
ben_ben=# select * from people where peoplesalary >= 10000;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |

        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |

        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |

        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |

        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |

        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |

        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |

        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |

       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |

       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |

       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |

       16 |            4 |      2 | 孙大勇     | 男        | 1990-02-21 00:00:00 |     13000.00 | 13698745214 | 南京          |               |

       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |

       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |

       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |

(18 rows)
```

3. 查询月薪大于等于 10000 的女员工信息(多条件)

```
ben_ben=# select * from people where peoplesalary >= 10000 and peoplesex = '女';
```

```
ben_ben=# select * from people where peoplesalary >= 10000 and peoplesex = '女';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtim
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------
--------
        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |
       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |
       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |
       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |
       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |
       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |
(9 rows)
```

4. 显示出出身年月在 1980-1-1 之后，而且月薪大于等于 10000 的女员工信息。

```
ben_ben=# select * from people where peoplesalary >= 10000 and peoplesex = '女' and peoplebirth > '1980-1-1';
```

```
ben_ben=# select * from people where peoplesalary >= 10000 and peoplesex = '女' and peoplebirth > '1980-1-1';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |
       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |
       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |
       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |
       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |
(7 rows)
```

5. 显示出月薪大于等于 15000 的员工，或者月薪大于等于 8000 的女员工信息

```
ben_ben=# select * from people where peoplesalary >= 10000 or ( peoplesex = '女' and peoplesalary >= 8000 );
```

```
ben_ben=# select * from people where peoplesalary >= 10000 or ( peoplesex = '女' and peoplesalary >= 8000 );
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |
        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |
        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |
        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |
        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |
        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |
       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |
       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |
       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |
       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |
       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |
       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |
       16 |            4 |      2 | 孙大勇     | 男        | 1990-02-21 00:00:00 |     13000.00 | 13698745214 | 南京          |               |
       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |
       21 |            1 |      3 | 王萌       | 女        | 1987-05-19 00:00:00 |      9500.00 | 13352197364 | 新疆          |               |
       22 |            2 |      1 | 陆子虹     | 女        | 1996-05-19 00:00:00 |      8500.00 | 13025457392 | 南京          |               |
       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |
```

```
(20 rows)
```

6. 查询月薪在 10000-20000 之间员工信息(多条件)

```
ben_ben=# select * from people where peoplesalary between 10000 and 20000;
```

```
ben_ben=# select * from people where peoplesalary between 10000 and 20000;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |

        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |

        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |

        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |

        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |

       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |

       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |

       16 |            4 |      2 | 孙大勇     | 男        | 1990-02-21 00:00:00 |     13000.00 | 13698745214 | 南京          |               |

       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |

       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |

       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |

(15 rows)
```

7. 查询出地址在北京或者上海的员工信息（用 or 和 in 两种方式）

方法一：or

```
ben_ben=# select * from people where peopleaddress = '北京' or peopleaddress = '上海';
```

方法二：in

```
ben_ben=# select * from people where peopleaddress in ('北京','上海');
```

```
ben_ben=# select * from people where peopleaddress = '北京' or peopleaddress = '上海';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

(3 rows)
```

```
ben_ben=# select * from people where peopleaddress in ('北京','上海');
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

(3 rows)
```

8. 查询所有员工信息(根据工资排序,降序排列) order by：排序 asc：正序 desc：倒序

```
ben_ben=# select * from people order by peoplesalary desc;
```

```
ben_ben=# select * from people order by peoplesalary desc;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |
        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |
        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |
       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |
        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |
       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |
       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |
       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |
       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |
       16 |            4 |      2 | 孙大勇     | 男        | 1990-02-21 00:00:00 |     13000.00 | 13698745214 | 南京          |               |
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |
       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |
        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |
       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |
        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |
       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |
        8 |            2 |      2 | 李小强     | 男        | 1989-04-20 00:00:00 |      9500.00 | 13335457412 | 山西          |               |
       21 |            1 |      3 | 王萌       | 女        | 1987-05-19 00:00:00 |      9500.00 | 13352197364 | 新疆          |               |
       19 |            5 |      2 | 周海涛     | 男        | 1992-10-11 00:00:00 |      9000.00 | 13987455214 | 天津          |               |
       22 |            2 |      1 | 陆子虹     | 女        | 1996-05-19 00:00:00 |      8500.00 | 13025457392 | 南京          |               |
       23 |            3 |      2 | 康军       | 男        | 1983-06-01 00:00:00 |      8500.00 | 13077778888 | 广西          |               |
       20 |            5 |      3 | 韦小宝     | 男        | 1984-09-10 00:00:00 |      6500.00 | 13254785965 | 宁夏          |               |
(24 rows)
```

9. 显示所有的员工信息，按照姓名的长度进行倒序排列

```
ben_ben=# select * from people order by length(peoplename) desc;
```

```
ben_ben=# select * from people order by length(peoplename) desc;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |
       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |
        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |
        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |
        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |
        8 |            2 |      2 | 李小强     | 男        | 1989-04-20 00:00:00 |      9500.00 | 13335457412 | 山西          |               |
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |
       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |
       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |
       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |
       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |
        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |
       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |
       19 |            5 |      2 | 周海涛     | 男        | 1992-10-11 00:00:00 |      9000.00 | 13987455214 | 天津          |               |
       20 |            5 |      3 | 韦小宝     | 男        | 1984-09-10 00:00:00 |      6500.00 | 13254785965 | 宁夏          |               |
       22 |            2 |      1 | 陆子虹     | 女        | 1996-05-19 00:00:00 |      8500.00 | 13025457392 | 南京          |               |
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |
       16 |            4 |      2 | 孙大勇     | 男        | 1990-02-21 00:00:00 |     13000.00 | 13698745214 | 南京          |               |
       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |
       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |
       21 |            1 |      3 | 王萌       | 女        | 1987-05-19 00:00:00 |      9500.00 | 13352197364 | 新疆          |               |
       23 |            3 |      2 | 康军       | 男        | 1983-06-01 00:00:00 |      8500.00 | 13077778888 | 广西          |               |
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |
(24 rows)
```

10. 查询工资最高的 5 个人的信息

```
ben_ben=# select * from people order by peoplesalary desc limit 5;
```

```
ben_ben=# select * from people order by peoplesalary desc limit 5;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |

        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |

        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |

(5 rows)
```

11. 查询工资最高的 2 个员工的信息

```
ben_ben=# select * from people order by peoplesalary desc limit 2;
```

```
ben_ben=# select * from people order by peoplesalary desc limit 2;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |

        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |

(2 rows)
```

12. 查询出地址没有填写的员工信息！（正常截图）

```
ben_ben=# select * from people where peopleaddress is null;
```

```
ben_ben=# select * from people where peopleaddress is null;
 peopleid | departmentid | rankid | peoplename | peoplesex | peoplebirth | peoplesalary | peoplephone | peopleaddress | peopleaddtime | mphone
----------+--------------+--------+------------+-----------+-------------+--------------+-------------+---------------+---------------+--------
(0 rows)
```

13. 查询出地址已经填写的员工信息

```
ben_ben=# select * from people where peopleaddress is not null;
```

```
ben_ben=# select * from people where peopleaddress is not null;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京市        |               |

        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |

        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |

        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |

        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |

        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |

        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |

        8 |            2 |      2 | 李小强     | 男        | 1989-04-20 00:00:00 |      9500.00 | 13335457412 | 山西          |               |

        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |

       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |

       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |

       16 |            4 |      2 | 孙大勇     | 男        | 1990-02-21 00:00:00 |     13000.00 | 13698745214 | 南京          |               |

       17 |            5 |      2 | 吴若欣     | 女        | 1995-02-21 00:00:00 |     14000.00 | 13985478512 | 湖南          |               |

       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |

       19 |            5 |      2 | 周海涛     | 男        | 1992-10-11 00:00:00 |      9000.00 | 13987455214 | 天津          |               |

       20 |            5 |      3 | 韦小宝     | 男        | 1984-09-10 00:00:00 |      6500.00 | 13254785965 | 宁夏          |               |

       21 |            1 |      3 | 王萌       | 女        | 1987-05-19 00:00:00 |      9500.00 | 13352197364 | 新疆          |               |

       22 |            2 |      1 | 陆子虹     | 女        | 1996-05-19 00:00:00 |      8500.00 | 13025457392 | 南京          |               |

       23 |            3 |      2 | 康军       | 男        | 1983-06-01 00:00:00 |      8500.00 | 13077778888 | 广西          |               |

       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |

(24 rows)
```

14. 查询所有的 80 后员工信息

```
ben_ben=# select * from people where extract(year from peoplebirth) between 1980 and 1989;
```

```
ben_ben=# select * from people where extract(year from peoplebirth) between 1980 and 1989;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        4 |            1 |      1 | 钱小旦     | 女        | 1988-08-08 00:00:00 |     13000.00 | 13985745871 | 山东          |               |

        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |

        8 |            2 |      2 | 李小强     | 男        | 1989-04-20 00:00:00 |      9500.00 | 13335457412 | 山西          |               |

        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |

       15 |            4 |      3 | 叶子飞云   | 女        | 1988-01-22 00:00:00 |     12000.00 | 13558745874 | 内蒙古        |               |

       20 |            5 |      3 | 韦小宝     | 男        | 1984-09-10 00:00:00 |      6500.00 | 13254785965 | 宁夏          |               |

       21 |            1 |      3 | 王萌       | 女        | 1987-05-19 00:00:00 |      9500.00 | 13352197364 | 新疆          |               |

       23 |            3 |      2 | 康军       | 男        | 1983-06-01 00:00:00 |      8500.00 | 13077778888 | 广西          |               |

(10 rows)
```

15. 查询年龄在 30-40 之间，并且工资在 15000-30000 之间的员工信息 (between and))

```
ben_ben=# select * from people where (extract(year from now()) - extract(year from peoplebirth)) between 30 and 40 and peoplesalary between 15000 and 30000;
```

```
ben_ben=# select * from people where (extract(year from now()) - extract(year from peoplebirth)) between 30 and 40 and peoplesalary between 150
00 and 30000;
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        5 |            2 |      1 | 张世同     | 男        | 1990-08-08 00:00:00 |     19000.00 | 13535987412 | 安徽          |               |

        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

(3 rows)
```

16. 查询出巨蟹座（即出生月份和日期是 6.22--7.22） 的员工信息

```
ben_ben=# select * from people where (extract(month from peoplebirth) = 6 and extract(day from peoplebirth) >= 22)
or (extract(month from peoplebirth) = 7 and extract(day from peoplebirth) <= 22);
```

```
ben_ben=# select * from people where (extract(month from peoplebirth) = 6 and extract(day from peoplebirth) >= 22)
or (extract(month from peoplebirth) = 7 and extract(day from peoplebirth) <= 22);
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |

        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |

(2 rows)
```

17. 查询工资比柳叶红云高的人！ （正常截图没有就没有）

```
ben_ben=# select * from people where peoplesalary > (select peoplesalary from people where peoplename = '柳叶红云');
```

```
ben_ben=# select * from people where peoplesalary > (select peoplesalary from people where peoplename = '柳叶红云');
 peopleid | departmentid | rankid | peoplename | peoplesex | peoplebirth | peoplesalary | peoplephone | peopleaddress | peopleaddtime | mphone
----------+--------------+--------+------------+-----------+-------------+--------------+-------------+---------------+---------------+--------
(0 rows)
```

18. 查询出和赵小明在同一个城市的人

```
ben_ben=# select * from people where peopleaddress = (select peopleaddress from people where peoplename = '赵小明') and peoplename <> '赵小明';
```

```
ben_ben=# select * from people where peopleaddress = (select peopleaddress from people where peoplename = '赵小明') and peoplename <> '赵小明';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        9 |            3 |      1 | 胡瑗瑗     | 女        | 1989-04-20 00:00:00 |     17500.00 | 13437100050 | 北京          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

(2 rows)
```

19. -查询姓刘的员工信息 % ：0 个字符，1 一个字符，多个字符！

```
ben_ben=# select * from people where peoplename like '刘%';
```

```
ben_ben=# select * from people where peoplename like '刘%';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |

       24 |            1 |      3 | 刘星宇     | 女        | 2004-09-07 00:00:00 |     18000.00 | 18717024225 | 重庆          |               |

(2 rows)
```

20. 查询姓刘的员工，名字是 2 个字 _:有且只有一个字符！（19、20 可能是一个结果，没关系）

```
ben_ben=# select * from people where peoplename like '刘_';
```

```
ben_ben=# select * from people where peoplename like '刘_';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
       18 |            5 |      1 | 刘婷       | 女        | 1996-02-21 00:00:00 |     14500.00 | 13778787874 | 湖北          |               |

(1 row)
```

21. 查询出名字最后一个字是“宝”，名字一共三个字的员工信息

```
ben_ben=# select * from people where peoplename like '__宝';
```

```
ben_ben=# select * from people where peoplename like '__宝';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
       20 |            5 |      3 | 韦小宝     | 男        | 1984-09-10 00:00:00 |      6500.00 | 13254785965 | 宁夏          |               |

(1 row)
```

22. 查询出电话号码开头 138 的员工信息

```
ben_ben=# select * from people where peoplephone like '138%';
```

```
ben_ben=# select * from people where peoplephone like '138%';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        6 |            2 |      3 | 委正大     | 男        | 1989-04-08 00:00:00 |     10000.00 | 13845789568 | 河北          |               |

        7 |            2 |      2 | 赵有飞     | 男        | 1995-04-08 00:00:00 |     11500.00 | 13878562568 | 浙江          |               |

       10 |            3 |      2 | 宋可晴     | 女        | 1987-12-20 00:00:00 |     16000.00 | 13889562354 | 北京          |               |

       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |

        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京          |               |

(5 rows)
```

23. 查询出电话号码开头 138 的员工信息,第 4 位可能是 7，可能 8 ，最后一个号码是 5

```
ben_ben=# select * from people where peoplephone like '1387%5' or peoplephone like '1388%5';
```

```
ben_ben=# select * from people where peoplephone like '1387%5' or peoplephone like '1388%5';
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
       13 |            4 |      1 | 曹德旺     | 男        | 1998-12-12 00:00:00 |     18500.00 | 13878523695 | 深圳          |               |

(1 row)
```

24. 统计职工表员工的总人数

```
ben_ben=# select count(*) from people;
```

```
ben_ben=# select count(*) from people;
 count
-------
    24
(1 row)
```

25. 求最大值 ,工资的最大值:统计单位员工薪资的最大值

```
ben_ben=# select max(peoplesalary) from people;
```

```
ben_ben=# select max(peoplesalary) from people;
   max
----------
 51000.00
(1 row)
```

26. 求最小值 ,工资的最小值:

```
ben_ben=# select min(peoplesalary) from people;
```

```
ben_ben=# select min(peoplesalary) from people;
   min
---------
 6500.00
(1 row)
```

27. 求工资总和:统计本单位月工资的总数

```
ben_ben=# select sum(peoplesalary) from people;
```

```
ben_ben=# select sum(peoplesalary) from people;
    sum
-----------
 385500.00
(1 row)
```

28. 求工资平均值:

```
ben_ben=# select avg(peoplesalary) from people;
```

```
ben_ben=# select avg(peoplesalary) from people;
        avg
--------------------
 16062.500000000000
(1 row)
```

29. 求数量，最大值，最小值，总和，平均值，在一行显示

```
ben_ben=# select count(*) 数量 , max(peoplesalary) 最大值 , min(peoplesalary) 最小值 , sum(peoplesalary) 总和 , avg(peoplesalary) 平均值 from people;
```

```
ben_ben=# select count(*) 数量 , max(peoplesalary) 最大值 , min(peoplesalary) 最小值 , sum(peoplesalary) 总和 , avg(peoplesalary) 平均值 from p
eople;
 数量 |  最大值   | 最小值  |   总和    |       平均值
------+----------+---------+-----------+--------------------
   24 | 51000.00 | 6500.00 | 385500.00 | 16062.500000000000
(1 row)
```

30. 查询出湖北地区的员工人数，总工资，最高工资，最低工资和平均工资

```
ben_ben=# select count(*) 员工人数 , max(peoplesalary) 最高工资 , min(peoplesalary) 最低工资 , sum(peoplesalary) 总工资 , avg(peoplesalary) 平均工资 fropeple group by peopleaddress having peopleaddress = '湖北';
```

```
ben_ben=# select count(*) 员工人数 , max(peoplesalary) 最高工资 , min(peoplesalary) 最低工资 , sum(peoplesalary) 总工资 , avg(peoplesalary) 平
均工资 fropeple group by peopleaddress having peopleaddress = '湖北';
 员工人数 | 最高工资 | 最低工资 |  总工资  |        平均工资
----------+----------+----------+----------+-----------------------
        1 | 14500.00 | 14500.00 | 14500.00 | 14500.0000000000000000
(1 row)
```

31. 计算出月薪在 10000 以上的男性员工的最大年龄，最小年龄和平均年龄

```
ben_ben=# select max(extract(year from now()) - extract(year from peoplebirth)) 最大年龄, min(extract(year from now()) - extract(year from peoplebirth)) 最小年龄 , avg(extract(year from now()) - extract(year from peoplebirth)) 平均年龄 from people where peoplesalary > 10000 group by peoplesex having peoplesex = '男';
```

```
ben_ben=# select max(extract(year from now()) - extract(year from peoplebirth)) 最大年龄, min(extract(year from now()) - extract(year from peop
lebirth)) 最小年龄 , avg(extract(year from now()) - extract(year from peoplebirth)) 平均年龄 from people where peoplesalary > 10000 group by pe
oplesex having peoplesex = '男';
 最大年龄 | 最小年龄 |     平均年龄
----------+----------+------------------
       57 |       27 | 40.8571428571429
(1 row)
```

32. 统计出所在地在“北京或上海”的所有女员工数量以及最大年龄，最小年龄和平均年龄。

```
ben_ben=# select count(*) 女员工数量 , max(extract(year from now()) - extract(year from peoplebirth)) 最大年龄 , min(extract(year from now()) - extract(year from peoplebirth)) 最小年龄 , avg(extract(year from now()) - extract(year from peoplebirth)) 平均年龄 from people where peopleaddress in ('北京','上海') and peoplesex = '女';
```

```
ben_ben=# select count(*) 女员工数量 , max(extract(year from now()) - extract(year from peoplebirth)) 最大年龄 , min(extract(year from now()) -
 extract(year from peoplebirth)) 最小年龄 , avg(extract(year from now()) - extract(year from peoplebirth)) 平均年龄 from people where peoplead
dress in ('北京','上海') and peoplesex = '女';
 女员工数量 | 最大年龄 | 最小年龄 | 平均年龄
------------+----------+----------+----------
          2 |       38 |       36 |       37
(1 row)
```

33. 求出年龄比平均年龄高的人员信息

```
ben_ben=# select * from people where extract(year from now()) - extract(year from peoplebirth) > (select avg(extract(year from now()) - extract
(year from peoplebirth)) from people);
```

```
ben_ben=# select * from people where extract(year from now()) - extract(year from peoplebirth) > (select avg(extract(year from now()) - extract
(year from peoplebirth)) from people);
 peopleid | departmentid | rankid | peoplename | peoplesex |     peoplebirth     | peoplesalary | peoplephone | peopleaddress | peopleaddtime |
 mphone
----------+--------------+--------+------------+-----------+---------------------+--------------+-------------+---------------+---------------+
--------
        2 |            1 |      1 | 司马相如   | 男        | 1968-07-09 00:00:00 |     41000.00 | 13554785452 | 成都          |               |

        3 |            1 |      1 | 周小顺     | 女        | 1972-07-09 00:00:00 |     26000.00 | 13256854578 | 甘肃          |               |

       11 |            3 |      3 | 肖飞       | 男        | 1981-11-11 00:00:00 |     10000.00 | 13385299632 | 河南          |               |

       12 |            3 |      1 | 张锐杰     | 男        | 1978-01-13 00:00:00 |     19000.00 | 13478545263 | 上海          |               |

       14 |            4 |      3 | 李萌       | 女        | 1968-11-22 00:00:00 |     10000.00 | 13698545841 | 广州          |               |

       20 |            5 |      3 | 韦小宝     | 男        | 1984-09-10 00:00:00 |      6500.00 | 13254785965 | 宁夏          |               |

       23 |            3 |      2 | 康军       | 男        | 1983-06-01 00:00:00 |      8500.00 | 13077778888 | 广西          |               |

        1 |            1 |      1 | 赵小明     | 男        | 1970-01-01 00:00:00 |     51000.00 | 13893330005 | 北京          |               |

(8 rows)
```

34. 创建数据库用户 jack

```
ben_ben=# create user jack identified by 'Jack123456';
```

```
ben_ben=# create user jack identified by 'Jack123456';
CREATE ROLE
```

35. 给用户 jack 授权对 DB_em 数据库的所有权限

```
ben_ben=# grant all on database ben_ben to jack;
```

```
ben_ben=# grant all on database ben_ben to jack;
GRANT
```

36. 给用户 jack 追加有创建角色的 CREATEROLE 权限

```
ben_ben=# alter user jack createrole;
```

```
ben_ben=# alter user jack createrole;
ALTER ROLE
```

37. 给用户 jack 追加 Sysadmin 的所有权限

```
ben_ben=# alter user jack with sysadmin;
```

```
ben_ben=# alter user jack with sysadmin;
ALTER ROLE
```

四. 实验过程中遇到的问题及解决方法：

主要的一个问题是我自主学习了基础的 mysql 数据库的 sql 语言，但是在部分语句上因为“方言”问题，导致语句不适配，例如 35 授权语句，在 mysql 中应该为：

grant all on ben_ben.* to jack;

但此处为：

grant all on datebase ben_ben to jack;

类似这样的方言问题还有几处，但所幸通过询问 ai 很快解决了问题。这也让我对于不同数据库 sql 语句的方言问题引起了重视。

五. 实验心得

本次实验主要练习的还是 DCL 和 DQL 语句，即数据库查询语句和数据库控制语句。以及聚合函数和其他几个函数的运用。在练习过程中巩固了我对于这些语句的掌握，并且，在循序渐进的练习中，我了解到许多之前没有涉及到的操作，例如 17、18 嵌套使用了 select 语句进行条件的限制。其实在做 18 题时，我就想到 where 条件中不能使用聚合函数的问题，但是又一时想不到如何不通过在 where 中使用聚合函数来实现条件的限制，不过我又知道聚合函数是一定要使用的，然后通过询问 ai 才恍然大悟，原来还可以这么写。

总的来说这次实验还是收获挺多的，我感觉最需要记忆的就是函数部分，那一块使用频率应该蛮大。